

Modal Logic: More Tableaux Examples

2026-02-23

Midterm

- The exam will be eight questions long.
- The questions will be like the ones on this week's quiz and the practice exam.
- It will be here, with answers to be done on paper.
- We'll find a separate room (to be announced) for folks needing accommodations.
- You'll get the cheat sheet, but otherwise it is closed book.

Quiz 6

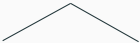
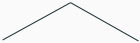
Question 1

$\Box(A \vee B) \rightarrow (\Diamond A \vee \Box B)$ is a theorem of $**KT**$

1.	$\mathbb{F} 1, \Box(A \vee B) \rightarrow (\Diamond A \vee \Box B)$ ✓	Assumption
2.	$\top 1, \Box(A \vee B)$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond A \vee \Box B$ ✓	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{F} 1, \Diamond A$	$\vee \mathbb{F} 3$
5.	$\mathbb{F} 1, \Box B$ ✓	$\vee \mathbb{F} 1$
6.	$\mathbb{F} 1.1, B$	$\Box \mathbb{F} 5$
7.	$\top 1.1, A \vee B$ ✓	$\Box \top 2$
$\swarrow \quad \searrow$		
8.	$\top 1.1, A \quad \top 1.1, B$	$\vee \top 7$
9.	$\mathbb{F} 1.1, A \quad \otimes$	$\Diamond \mathbb{F} 4$
	$\otimes$	

Question 2

$\Box(A \rightarrow B) \rightarrow (A \rightarrow \Box B)$ is not a theorem of $**KT**$

1.	$\mathbb{F} 1, \Box(A \rightarrow B) \rightarrow (A \rightarrow \Box B)$ ✓	Assumption
2.	$\mathbb{T} 1, \Box(A \rightarrow B)$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, A \rightarrow \Box B$ ✓	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, A$	$\rightarrow \mathbb{F} 3$
5.	$\mathbb{F} 1, \Box B$	$\rightarrow \mathbb{F} 3$
6.	$\mathbb{T} 1, A \rightarrow B$ ✓	$\mathbb{T} \Box 2$
		
7.	$\mathbb{F} 1, A$ $\mathbb{T} 1, B$	$\rightarrow \mathbb{T} 6$
8.	⊗ $\mathbb{F} 1.1, B$	$\Box \mathbb{F} 5$
9.	$\mathbb{T} 1.1, A \rightarrow B$ ✓	$\Box \mathbb{T} 2$
		
10.	$\mathbb{F} 1.1, A$ $\mathbb{T} 1.1, B$	$\rightarrow \mathbb{T} 9$
	⊗	

Question 2

The middle branch stays open, and the open-ended instruction at line 2 has been applied everywhere that's needed.

The counterexample is where A and B are both true at w_1 and both false at $w_{1.1}$.

Question 3

$\Box A \rightarrow (A \wedge \Box \Box A)$ is not a theorem of $**K4**$

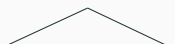
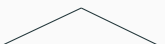
1.	$\mathbb{F} 1, \Box A \rightarrow (A \wedge \Box \Box A)$ ✓	Assumption
2.	$\mathbb{T} 1, \Box A$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, A \wedge \Box \Box A$ ✓	$\rightarrow \mathbb{F} 1$
$\swarrow \quad \searrow$		
4.	$\mathbb{F} 1, A$ $\mathbb{F} 1, \Box \Box A$ ✓	$\wedge \mathbb{F} 3$
5.	$\mathbb{F} 1.1, \Box A$	$\Box \mathbb{F} 4$
6.	$\mathbb{T} 1.1, \Box A$	$4 \Box 2$
	⊗	

Question 3

The left hand branch stays open. The counterexample is the somewhat degenerate model where w_1 can't access anything, and has A false.

Question 4

$\Diamond(A \rightarrow \Diamond B) \rightarrow (\Box A \rightarrow \Diamond(A \wedge B))$ is a theorem of ***K4***

1.	$\mathbb{F} 1, \Diamond(A \rightarrow \Diamond B) \rightarrow (\Box A \rightarrow \Diamond(A \wedge B))$ ✓	Assumption
2.	$\mathbb{T} 1, \Diamond(A \rightarrow \Diamond B)$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box A \rightarrow \Diamond(A \wedge B)$ ✓	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Box A$	$\rightarrow \mathbb{F} 3$
5.	$\mathbb{F} 1, \Diamond(A \wedge B)$	$\rightarrow \mathbb{F} 3$
6.	$\mathbb{T} 1.1, A \rightarrow \Diamond B$ ✓	$\Diamond \mathbb{T} 2$
7.	$\mathbb{T} 1.1, A$	$\Box \mathbb{T} 4$
8.	$\mathbb{F} 1.1, A \wedge B$	$\Diamond \mathbb{F} 5$
		
9.	$\mathbb{F} 1.1, A \quad \mathbb{T} 1.1, \Diamond B$ ✓	$\rightarrow \mathbb{T} 6$
10.	$\otimes \quad \mathbb{T} 1.1, \Box A$	$4 \Box 4$
11.	$\mathbb{F} 1.1, \Diamond(A \wedge B)$	$4 \Diamond 5$
12.	$\mathbb{T} 1.1.1, B$	$\Diamond \mathbb{T} 9$
13.	$\mathbb{T} 1.1.1, A$	$\Box \mathbb{T} 10$
14.	$\mathbb{F} 1.1.1, A \wedge B$ ✓	$\Diamond \mathbb{F} 11$
		
15.	$\mathbb{F} 1.1.1, A \quad \mathbb{F} 1.1.1, B$	$\wedge \mathbb{F} 14$

Question 5

$\Diamond(A \wedge B) \rightarrow \Box\Diamond(C \rightarrow B)$ is not a theorem of $**S4**$

1.	$\mathbb{F} 1, \Diamond(A \wedge B) \rightarrow \Box\Diamond(C \rightarrow B)$ ✓	Assumption
2.	$\mathbb{T} 1, \Diamond(A \wedge B)$ ✓	$\rightarrow\mathbb{F} 1$
3.	$\mathbb{F} 1, \Box\Diamond(C \rightarrow B)$ ✓	$\rightarrow\mathbb{F} 1$
4.	$\mathbb{T} 1.1, A \wedge B$ ✓	$\Diamond\mathbb{T} 2$
5.	$\mathbb{T} 1.1, A$	$\wedge\mathbb{T} 4$
6.	$\mathbb{T} 1.1, B$	$\wedge\mathbb{T} 4$
7.	$\mathbb{F} 1.2, \Diamond(C \rightarrow B)$	$\Box\mathbb{F} 3$
8.	$\mathbb{F} 1.2, C \rightarrow B$ ✓	$\mathbb{T}\Diamond 7$
9.	$\mathbb{T} 1.2, C$	$\rightarrow\mathbb{F} 8$
10.	$\mathbb{F} 1.2, B$	$\rightarrow\mathbb{F} 8$

Question 5

The trunk stays open, and we've applied the open-ended rule everywhere it needs to be applied.

The countermodel has w_1 , which can access $w_{1.1}$ and $w_{1.2}$. At $w_{1.1}$ both A and B are true. At $w_{1.2}$, C is true and B is false.

Any way of filling in the other values will still create a counterexample.

Question 6

$\Diamond\Diamond A \rightarrow (\Diamond A \wedge \Diamond\Diamond\Diamond A)$ is a theorem of $\ast\ast S4 \ast\ast$

1.	$\mathbb{F} 1, \Diamond\Diamond A \rightarrow (\Diamond A \wedge \Diamond\Diamond\Diamond A)$ ✓	Assumption
2.	$\top 1, \Diamond\Diamond A$ ✓	$\rightarrow\mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond A \wedge \Diamond\Diamond\Diamond A$ ✓	$\rightarrow\mathbb{F} 1$
4.	$\top 1.1, \Diamond A$	$\Diamond\top 2$
$\begin{array}{c} \swarrow \quad \searrow \\ \mathbb{F} 1, \Diamond A \quad \mathbb{F} 1, \Diamond\Diamond\Diamond A \end{array}$		$\wedge\mathbb{F} 3$
5.		
6.	$\mathbb{F} 1.1, \Diamond A$	$4\Diamond 5$
7.	⊗	$\top\Diamond 6$
	$\mathbb{F} 1, \Diamond\Diamond A$	
	⊗	

There are a lot of ways to close this one; this is the quickest I found but it's definitely not the only one.

Question 7

$\Diamond \Box A \rightarrow \Diamond A$ is a theorem of $**KTB**$

1.	$\mathbb{F} 1, \Diamond \Box A \rightarrow \Diamond A$ ✓	Assumption
2.	$\mathbb{T} 1, \Diamond \Box A$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond A$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1.1, \Box A$	$\Diamond \mathbb{T} 2$
5.	$\mathbb{T} 1, A$	$\Box \mathbb{F} 4$
6.	$\mathbb{F} 1, A$	$\mathbb{T} \Diamond 3$
	⊗	

Question 8

$(\Box A \wedge \Box B) \rightarrow \Box\Box(A \rightarrow B)$ is a not theorem of $**KTB**$

1.	$\mathbb{F} 1, (\Box A \wedge \Box B) \rightarrow \Box \Box (A \rightarrow B)$ ✓	Assumption
2.	$\mathbb{T} 1, \Box A \wedge \Box B$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box \Box (A \rightarrow B)$ ✓	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Box A$	$\wedge \mathbb{T} 2$
5.	$\mathbb{T} 1, \Box B$	$\wedge \mathbb{T} 2$
6.	$\mathbb{T} 1, A$	$\mathbb{T} \Box 4$
7.	$\mathbb{T} 1, B$	$\mathbb{T} \Box 5$
8.	$\mathbb{F} 1.1, \Box (A \rightarrow B)$ ✓	$\Box \mathbb{F} 3$
9.	$\mathbb{T} 1.1, A$	$\Box \mathbb{T} 6$
10.	$\mathbb{T} 1.1, B$	$\Box \mathbb{T} 7$
11.	$\mathbb{F} 1.1.1, A \rightarrow B$ ✓	$\Box \mathbb{F} 8$
12.	$\mathbb{T} 1.1.1, A$	$\rightarrow \mathbb{F} 11$
13.	$\mathbb{F} 1.1.1, B$	$\rightarrow \mathbb{F} 11$

Question 8

The tableau is open.

The countermodel takes a bit more work to describe than the previous ones.

There are three worlds, w_1 , $w_{1.1}$ and $w_{1.1.1}$.

Every world can access every world *except*, w_1 and $w_{1.1.1}$ can access each other.

By the naming convention, $w_1 R w_{1.1}$ and $w_{1.1} R w_{1.1.1}$; adding reflexivity and symmetry get us the rest of the relations.

A is true at all three worlds, B is true at the first and second, but not the third.

Practice Exam

Question 1

$\Box(\Box p \rightarrow (q \vee r)) \vee \Box(\Box q \rightarrow p)$ in S5

1.	$\mathbb{F} 1, \Box(\Box p \rightarrow (q \vee r)) \vee \Box(\Box q \rightarrow p)$ ✓	Assumption
2.	$\mathbb{F} 1, \Box(\Box p \rightarrow (q \vee r))$ ✓	$\vee\mathbb{F} 1$
3.	$\mathbb{F} 1, \Box(\Box q \rightarrow p)$ ✓	$\vee\mathbb{F} 1$
4.	$\mathbb{F} 2, \Box p \rightarrow (q \vee r)$ ✓	$\Box\mathbb{F} 2$
5.	$\mathbb{T} 2, \Box p$	$\rightarrow\mathbb{F} 4$
6.	$\mathbb{F} 2, q \vee r$ ✓	$\rightarrow\mathbb{F} 4$
7.	$\mathbb{F} 2, q$	$\vee\mathbb{F} 6$
8.	$\mathbb{F} 2, r$	$\vee\mathbb{F} 6$
9.	$\mathbb{F} 3, \Box q \rightarrow p$	$\Box\mathbb{F} 3$
10.	$\mathbb{T} 3, \Box q$	$\rightarrow\mathbb{F} 9$
11.	$\mathbb{F} 3, p$	$\rightarrow\mathbb{F} 9$
12.	$\mathbb{T} 3, p$	$\Box\mathbb{T} 5$
	⊗	

The only branch closes, so this is **valid** in **S5**.

Question 2

$\Diamond \Box (p \rightarrow q) \rightarrow (\Box p \rightarrow \Diamond q)$ in S5

1.	$\mathbb{F} 1, \Diamond \Box (p \rightarrow q) \rightarrow (\Box p \rightarrow \Diamond q) \checkmark$	Assumption
2.	$\mathbb{T} 1, \Diamond \Box (p \rightarrow q) \checkmark$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box p \rightarrow \Diamond q \checkmark$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Box p$	$\rightarrow \mathbb{F} 3$
5.	$\mathbb{F} 1, \Diamond q$	$\rightarrow \mathbb{F} 3$
6.	$\mathbb{T} 2, \Box (p \rightarrow q)$	$\Diamond \mathbb{T} 2$
7.	$\mathbb{T} 2, p \rightarrow q \checkmark$	$\Box \mathbb{T} 2$
8.	$\mathbb{T} 2, p$	$\Box \mathbb{T} 4$
9.	$\mathbb{F} 2, q$	$\Diamond \mathbb{F} 5$
$ \begin{array}{c} \swarrow \quad \searrow \\ \mathbb{F} 2, p \quad \mathbb{T} 2, q \end{array} $		
10.	$ \begin{array}{cc} \mathbb{F} 2, p & \mathbb{T} 2, q \\ \otimes & \otimes \end{array} $	$\rightarrow \mathbb{T} 7$

Both branches close, so this is **valid** in S5.

Question 3

$\Box(p \vee q) \rightarrow (\Diamond(\neg q \vee r) \rightarrow \Diamond(p \vee r))$ in S5

1.	$\mathbb{F} 1, \Box(p \vee q) \rightarrow (\Diamond(\neg q \vee r) \rightarrow \Diamond(p \vee r))$ ✓	Assumption
2.	$\mathbb{T} 1, \Box(p \vee q)$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond(\neg q \vee r) \rightarrow \Diamond(p \vee r)$ ✓	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Diamond(\neg q \vee r)$ ✓	$\rightarrow \mathbb{F} 3$
5.	$\mathbb{F} 1, \Diamond(p \vee r)$	$\rightarrow \mathbb{F} 3$
6.	$\mathbb{T} 2, \neg q \vee r$ ✓	$\Diamond \mathbb{T} 4$
7.	$\mathbb{T} 2, p \vee q$ ✓	$\Box \mathbb{T} 1$
8.	$\mathbb{F} 2, p \vee r$ ✓	$\Diamond \mathbb{F} 5$
9.	$\mathbb{F} 2, p$	$\vee \mathbb{F} 8$
10.	$\mathbb{F} 2, r$	$\vee \mathbb{F} 8$
11.	$ \begin{array}{cc} \mathbb{T} 2, p & \mathbb{T} 2, q \\ \text{⊗} & \mathbb{T} 2, q \end{array} $	$\vee \mathbb{T} 7$
12.	$ \begin{array}{ccc} & \mathbb{T} 2, \neg q \text{ ✓} & \mathbb{T} 2, r \\ & \mathbb{F} 2, q & \text{⊗} \end{array} $	$\vee \mathbb{T} 6$
13.	⊗	$\neg \mathbb{T} 12$

Question 4

$$(\neg \Diamond p \vee \neg \Diamond \Diamond p) \rightarrow \neg \Diamond \Diamond \Diamond p \text{ in S5}$$

1.	$\mathbb{F} 1, (\neg\Diamond p \vee \neg\Diamond\Diamond p) \rightarrow \neg\Diamond\Diamond\Diamond p$ ✓	Assumption
2.	$\mathbb{T} 1, \neg\Diamond p \vee \neg\Diamond\Diamond p$ ✓	$\rightarrow\mathbb{F} 1$
3.	$\mathbb{F} 1, \neg\Diamond\Diamond\Diamond p$ ✓	$\rightarrow\mathbb{F} 1$
4.	$\mathbb{T} 1, \Diamond\Diamond\Diamond p$ ✓	$\neg\mathbb{F} 3$
5.	$\mathbb{T} 2, \Diamond\Diamond p$ ✓	$\Diamond\mathbb{T} 4$
6.	$\mathbb{T} 3, \Diamond p$ ✓	$\Diamond\mathbb{T} 5$
7.	$\mathbb{T} 4, p$	$\Diamond\mathbb{T} 6$
$\swarrow \quad \searrow$		
8.	$\mathbb{T} 1, \neg\Diamond p$ ✓ $\mathbb{T} 1, \neg\Diamond\Diamond p$ ✓	$\vee\mathbb{T} 2$
9.	$\mathbb{F} 1, \Diamond p$	$\neg\mathbb{T} 8$
10.	$\mathbb{F} 4, p$	$\Diamond\mathbb{F} 9$
11.	$\otimes$ $\mathbb{F} 1, \Diamond\Diamond p$	$\neg\mathbb{T} 8$
12.		$\Diamond\mathbb{F} 11$
	$\otimes$	

Both branches close, so this is **valid** in **S5**. There are many ways to close 30 / 44

Question 5

$$(\Diamond p \wedge \Box q) \rightarrow \Diamond q \text{ in K}$$

1.	$\mathbb{F} 1, (\Diamond p \wedge \Box q) \rightarrow \Diamond q$ ✓	Assumption
2.	$\mathbb{T} 1, \Diamond p \wedge \Box q$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond q$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Diamond p$ ✓	$\wedge \mathbb{T} 2$
5.	$\mathbb{T} 1, \Box q$	$\wedge \mathbb{T} 2$
6.	$\mathbb{T} 1.1, p$	$\Diamond \mathbb{T} 4$
7.	$\mathbb{F} 1.1, q$	$\Diamond \mathbb{F} 3$
8.	$\mathbb{T} 1.1, q$	$\Box \mathbb{T} 5$



The trunk closes, so it is **valid** in **K**. The role of $\Diamond p$ in this one is weird.

Question 6

$$\Box(p \wedge \neg p) \vee \Diamond(q \vee \neg q) \text{ in K}$$

1.	$\models 1, \Box(p \wedge \neg p) \vee \Diamond(q \vee \neg q)$ ✓	Assumption
2.	$\models 1, \Box(p \wedge \neg p)$ ✓	$\vee\mathcal{E}$ 1
3.	$\models 1, \Diamond(q \vee \neg q)$	$\vee\mathcal{E}$ 1
4.	$\models 1.1, p \wedge \neg p$	$\Box\mathcal{E}$ 2
5.	$\models 1.1, q \vee \neg q$ ✓	$\Diamond\mathcal{E}$ 3
6.	$\models 1.1, q$	$\wedge\mathcal{E}$ 5
7.	$\models 1.1, \neg q$ ✓	$\vee\mathcal{E}$ 5
8.	$\top 1.1, q$	$\neg\mathcal{E}$ 7



This is also valid, and it's also wrong for the same reason.

Question 6

Not at all on the test digression.

This example shows that **K** has a very weird property. It is not **Halldén-complete**.

A logician, Sören Halldén, in the 1950s noticed that some logics have the odd property that $X \vee Y$ is a theorem even though (a) neither X nor Y is, and (b) X and Y have no variables in common.

We've come to call such logics Halldén-incomplete, and the opposite property Halldén-completeness.

In general, the logics that are interesting and get used in philosophical applications are Halldén-complete.

Question 7

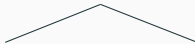
$$\Box \neg (p \vee \neg q) \rightarrow \Box p \text{ in K}$$

1.	$\mathbb{F} 1, \Box \neg(p \vee \neg q) \rightarrow \Box p$ ✓	Assumption
2.	$\mathbb{T} 1, \Box \neg(p \vee \neg q)$	$\rightarrow \mathbb{T} 1$
3.	$\mathbb{F} 1, \Box p$ ✓	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{F} 1.1, p$	$\Box \mathbb{F} 3$
5.	$\mathbb{T} 1.1, \neg(p \vee \neg q)$ ✓	$\Box \mathbb{T} 2$
6.	$\mathbb{F} 1.1, p \vee \neg q$ ✓	$\neg \mathbb{T} 5$
7.	$\mathbb{F} 1.1, p$	$\vee \mathbb{F} 6$
8.	$\mathbb{F} 1.1, \neg q$ ✓	$\vee \mathbb{F} 6$
9.	$\mathbb{T} 1.1, q$	$\neg \mathbb{F} 8$

The tree is open, so this is not a theorem, it is **invalid** in **K**. The countermodel has two worlds: w_1 and $w_1.1$, with $w_1 R w_{1.1}$. At $w_{1.1}$, p is false and q is true.

Question 8

$$\Box(p \rightarrow (q \wedge r)) \rightarrow (\Diamond\Diamond p \rightarrow \Diamond(q \vee s)) \text{ in K}$$

1.	$\mathbb{F} 1, \Box(p \rightarrow (q \wedge r)) \rightarrow (\Diamond\Diamond p \rightarrow \Diamond(q \vee s))$ ✓	Assumption
2.	$\mathbb{T} 1, \Box(p \rightarrow (q \wedge r))$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond\Diamond p \rightarrow \Diamond(q \vee s)$ ✓	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Diamond\Diamond p$ ✓	$\rightarrow \mathbb{F} 3$
5.	$\mathbb{F} 1, \Diamond(q \vee s)$	$\rightarrow \mathbb{F} 3$
6.	$\mathbb{T} 1.1, \Diamond p$ ✓	$\Diamond \mathbb{T} 4$
7.	$\mathbb{T} 1.1, \Box(p \rightarrow (q \wedge r))$	$4 \Box 2$
8.	$\mathbb{F} 1.1, \Diamond(q \vee s)$	$4 \Diamond 5$
9.	$\mathbb{T} 1.1.1, p$	$\Diamond \mathbb{T} 6$
10.	$\mathbb{T} 1.1.1, p \rightarrow (q \wedge r)$	$\Box \mathbb{T} 7$
		
11.	$\mathbb{F} 1.1.1, p$	$\rightarrow \mathbb{T} 10$
12.	$\otimes$ $\mathbb{T} 1.1.1, q$ ✓	$\wedge \mathbb{T} 11$
13.	$\mathbb{T} 1.1.1, r$	$\wedge \mathbb{T} 11$
14.	$\mathbb{F} 1.1.1, q \vee s$	$\Diamond \mathbb{F} 8$
15.	$\mathbb{F} 1.1.1, q$	$\vee \mathbb{F} 14$

Question 9

$$(\Diamond \Box p \wedge \Diamond \Box q) \rightarrow \Diamond(p \wedge q) \text{ in K}$$

1.	$\mathbb{F} 1, (p \wedge \Diamond \Box p) \rightarrow \Box p$ ✓	Assumption
2.	$\mathbb{T} 1, p \wedge \Diamond \Box p$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box p$ ✓	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, p$	$\wedge \mathbb{T} 2$
5.	$\mathbb{T} 1, \Diamond \Box p$ ✓	$\wedge \mathbb{T} 2$
6.	$\mathbb{F} 1.1, p$	$\Box \mathbb{F} 3$
7.	$\mathbb{T} 1.2, \Box p$	$\Diamond \mathbb{T} 5$
8.	$\mathbb{T} 1.2, p$	$\mathbb{T} \Box 7$

Question 9

This is open, so it isn't valid.

The countermodel has three worlds, w_1 , $w_{1.1}$ and $w_{1.2}$.

w_1 can access everything, the other two can only access themselves. And p is true at w_1 and $w_{1.2}$, but false at $w_{1.1}$.

Question 10

$$\Box\Diamond\Box\Diamond p \rightarrow \Box\Diamond p \text{ in K}$$

1.	$\mathbb{F} 1, \Diamond \Box \Diamond \Box p \rightarrow \Diamond \Box p$ ✓	Assumption
2.	$\mathbb{T} 1, \Diamond \Box \Diamond \Box p$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond \Box p$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1.1, \Box \Diamond \Box p$	$\Diamond \mathbb{T} 2$
5.	$\mathbb{F} 1.1, \Diamond \Box p$	$4 \Diamond 3$
6.	$\mathbb{T} 1.1, \Diamond \Box p$	$\mathbb{T} \Box 4$



The trunk closes, so this is **valid**. There are *a lot* of ways to do this one; this is one of the more direct.